

Cubic-Rate Irrationality of Reciprocal Sums

Erdős Problem #243

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ABSTRACT

We prove irrationality of the reciprocal sum when strictly increasing positive integers satisfy $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$. The proof extracts a cubic polynomial from an integer tail numerator, then uses a square in a cubic field to restrict its coefficients. Congruences modulo seven exclude the remaining possibilities. The unrestricted Erdős–Graham question remains open.

1 Introduction

The Sylvester recurrence $a_{n+1} = a_n^2 - a_n + 1$ gives a rational reciprocal sum by telescoping:

$$\frac{1}{a_n - 1} = \frac{1}{a_n} + \frac{1}{a_{n+1} - 1}.$$

For an increasing sequence of positive integers that eventually satisfies this recurrence, the tail sum at every sufficiently large index is $1/(a_n - 1)$. Erdős and Graham asked whether every rational reciprocal sum with $a_{n+1} \sim a_n^2$ arises in this way.

Problem 1.1 (Erdős Problem 243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

Erdős and Graham [1, p. 64] asked this question; see also Erdős [2, p. 105] and Bloom’s catalogue [5]. We prove the following irrationality result for a restricted class of sequences with $a_{n+1} \sim a_n^2$.

Theorem 1.2 (cubic-rate irrationality). *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Under the eventual Sylvester recurrence, $a_n^2/a_{n+1} - 1 = O(1/a_n)$, which is much smaller than $3/n$. Theorem 1.2 therefore treats a restricted case of the problem by excluding rationality altogether. The general quadratic limit $a_{n+1} \sim a_n^2$ remains insufficient for any argument given here.

For example, set $a_1 = 8$ and

$$a_{n+1} = \left\lceil \frac{na_n^2}{n+3} \right\rceil.$$

This gives 8, 16, 103, 5305, From $a_{n+1} \geq a_n^2/4$ we obtain $a_n \geq 4 \cdot 2^{2^{n-1}}$ by induction. The error introduced by the ceiling is less than one, and hence

$$0 \leq 1 + \frac{3}{n} - \frac{a_n^2}{a_{n+1}} < \frac{16}{a_n^2} = o(n^{-3}).$$

The theorem applies to this explicitly defined sequence.

If the sum is rational, clearing denominators gives positive integer tail numerators C_n . Their growth is cubic, so we compare them with $F_n = n(n+1)(n+2)$, for which $F_{n+1}/F_n = 1+3/n$. The ratio estimate gives $C_n = AF_n + o(n)$. Subtracting the recurrences for C_n and AF_n then shows that the first difference of the error tends to zero. It follows that $\Delta^4 C_n \rightarrow 0$. Since these fourth differences are integers, they eventually vanish, and $C_n = AF_n + B$.

We next use the denominator recurrence to constrain this polynomial. It gives a square condition at every root of a cubic modulo almost every prime. Chebotarev's theorem implies that the corresponding element of the cubic number field is a square. Trace identities leave two possible polynomials; four consecutive terms modulo seven exclude both. The proof in Section 3 follows this order.

The proof uses the integer tail numerators of Koizumi [4, Lemma 4]. Section 2 introduces these numerators, and Section 3 proves the theorem. The long record treats bounded increases, summable relative increases and weighted crossings of new maxima as separate criteria for the Sylvester recurrence. Those criteria do not resolve the general question.

2 Tail numerators

For the Sylvester sequence 2, 3, 7, 43, ..., whose reciprocal sum is 1, the successive tails are $1, 1/2, 1/6, 1/42, \dots$. Multiplication by the products $1, 2, 6, 42, \dots$ leaves the constant integer numerator 1. For an arbitrary rational reciprocal sum, we also multiply by a denominator of the whole sum. The resulting integer numerators need not be constant.

Assume now that a_n are strictly increasing positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n = p/q$ with positive integers p, q . All indices in this construction start at 1. Put

$$x_n = \sum_{k \geq n} \frac{1}{a_k}, \quad P_n = \prod_{1 \leq j < n} a_j, \quad D_n = qP_n, \quad C_n = D_n x_n.$$

Here $P_1 = 1$, $D_1 = q$ and $C_1 = p$. Since every a_k with $k < n$ divides P_n , subtracting the finite prefix from p/q gives

$$C_n = pP_n - q \sum_{1 \leq k < n} \frac{P_n}{a_k} \in \mathbb{N}_{>0}.$$

The integer is positive because it equals the positive tail multiplied by D_n . Removing the term $1/a_n$ from the tail gives

$$C_{n+1} = a_n C_n - D_n, \quad D_{n+1} = a_n D_n. \quad (1)$$

Koizumi constructs these coordinates in Lemma 4 [4, pp. 11–12]. On a pseudo-greedy tail his c_n, d_n, e_n correspond to $C_n, D_n, D_n - (a_n - 1)C_n$. If the tail is restarted with its value in lowest terms, both coordinates are divided by the same fixed integer. We retain the unreduced fraction C_n/D_n to keep the denominator update multiplicative. In the cubic proof we first extract a polynomial for this C_n , then divide out the eventual fixed gcd in Section 3.2.

For large n , $a_{n+1} \geq a_n^2/2 \geq 2a_n$. The terms after $1/a_{n+1}$ sum to at most $2/a_{n+2} \leq 4/a_{n+1}^2$, so

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}), \quad \frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n). \quad (2)$$

For the second estimate, the first gives $a_n x_n = 1 + O(1/a_n)$. Use this in

$$\frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} \frac{a_{n+1} x_{n+1}}{a_n x_n}.$$

The last factor is $1 + O(1/a_n)$, and $a_n^2/a_{n+1} \rightarrow 1$, giving the stated additive error. Quadratic growth makes $1/a_n = o(n^{-k})$ for every fixed k , so this error preserves the precision required in the cubic theorem. We keep the original index n throughout that proof. Although an eventual recurrence survives removal of a finite prefix, the expression $1 + 3/n + o(n^{-3})$ changes its lower-order terms under an index shift.

3 Proof of cubic-rate irrationality

Suppose, towards a contradiction, that the reciprocal sum in Theorem 1.2 is rational. Construct C_n, D_n as in Section 2. Equation (2) transfers the rate without changing its precision:

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}). \quad (3)$$

We first show that C_n is eventually a polynomial, using (3) and integrality. We then apply (1) to restrict its coefficients.

3.1 An eventual cubic polynomial

The factor $1 + 3/n$ suggests the cubic $F_n = n(n+1)(n+2)$, since $F_{n+1}/F_n = 1 + 3/n$. Write (3) as $C_{n+1} = (1 + 3/n + \varepsilon_n)C_n$, where $\varepsilon_n = o(n^{-3})$. Dividing out the exact model gives

$$z_n = \frac{C_n}{F_n}, \quad \frac{z_{n+1}}{z_n} = 1 + \frac{\varepsilon_n}{1 + 3/n}.$$

The logarithmic increments are absolutely summable, with tails $o(n^{-2})$. Hence z_n tends to some $A > 0$ and $z_n - A = o(n^{-2})$. Multiplying back by F_n yields

$$C_n = AF_n + \delta_n, \quad \delta_n = o(n).$$

The estimate $\delta_n = o(n)$ alone does not control its differences. Instead, subtracting the recurrences for C_n and AF_n gives

$$\Delta\delta_n = \frac{3}{n}\delta_n + \varepsilon_n C_n = o(1), \quad \Delta h_n = h_{n+1} - h_n.$$

Here $C_n = O(n^3)$ makes the second term $o(1)$. Three more differences still tend to zero and kill the cubic model:

$$\Delta^4 C_n = \Delta^3(\Delta\delta_n) \rightarrow 0.$$

The left side is an integer, so it vanishes for all sufficiently large n . Newton interpolation therefore expresses C_n on that tail as a rational polynomial of degree at most three. Its difference from AF_n is $o(n)$, which permits only a constant term:

$$C_n = An(n+1)(n+2) + B, \quad A \in \mathbb{Q}_{>0}, \quad B \in \mathbb{Q}. \quad (4)$$

The precision matters at the subtraction step. The integer sequence $F_n + (-1)^n$ has ratio $1 + 3/n + O(n^{-3})$ and never agrees eventually with a polynomial; its error has an oscillating first difference. This example concerns polynomial extraction alone. It is not asserted to satisfy the denominator recurrence.

3.2 Reduction and the constant term

We now use the denominator recurrence. Both updates preserve common divisors, so $G_n = \gcd(C_n, D_n)$ forms a divisibility chain: $G_n \mid G_{n+1}$. For every sufficiently late n , it divides four consecutive values of C_n , and hence their third difference $6A$. Unlike the vanishing fourth difference used above, this positive third difference bounds the gcd. Thus G_n stabilises at a positive integer g .

We can therefore reduce all sufficiently late fractions by the same factor: put $u_n = C_n/g$ and $v_n = D_n/g$. The recurrences persist after this division. The following observation describes the resulting coprimality conditions.

Proposition 3.1 (persistent coprimality). *Let a_n, u_n, v_n be integer sequences with $u_n > 0$, $v_n \geq 0$ and $\gcd(u_n, v_n) = 1$ satisfying*

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n.$$

Then $\gcd(a_n, v_n) = 1$. The a_n are pairwise coprime, and $\gcd(a_i, u_t) = 1$ whenever $i < t$.

Proof. A prime dividing both a_n and v_n would divide u_{n+1} and v_{n+1} , contrary to reducedness. For $i < t$, the factor a_i divides v_t . Reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. $\square$

In the cubic case, adjacent numerators are also coprime, since $\gcd(u_n, u_{n+1}) = \gcd(u_n, v_n) = 1$. Write the polynomial for u_n as

$$Q(n) = \frac{m}{6}n(n+1)(n+2) + c. \quad (5)$$

The third difference $m = 6A/g$ is a positive integer. Since $n(n+1)(n+2)/6 = \binom{n+2}{3}$ is an integer, any sufficiently late value $u_n = Q(n)$ gives

$$c = u_n - m \binom{n+2}{3} \in \mathbb{Z}.$$

Thus Q is integer-valued, although its coefficients need not all be integers. If a prime ℓ divides c , choose an arbitrarily late $n \equiv -1 \pmod{6\ell}$. Both $Q(n)$ and $Q(n+1)$ would be divisible by ℓ , contradicting adjacent coprimality. This also excludes $c = 0$. Thus

$$m \in \mathbb{N}_{>0}, \quad c \in \{1, -1\}.$$

We also need irreducibility. The multipliers $a_n > 1$ on the reduced tail are pairwise coprime, so infinitely many distinct primes occur among them. Once a prime divides one such a_n , it divides every later v_t and no later u_t . If the cubic Q had a rational root, that root would reduce to a root modulo all but finitely many primes. Choose one of the persistent primes outside those exceptions; a sufficiently late index in the root's residue class would have $\ell \mid Q(t) = u_t$, a contradiction. A cubic is reducible over $\mathbb{Q}$ only if it has a rational root. Hence Q is irreducible.

Set $\kappa = m/6$, $\eta = 6c/m$, and

$$f(T) = T^3 - T + \eta, \quad Q(n) = \kappa f(n+1).$$

3.3 A square forced by a zero numerator

A zero numerator modulo a prime eliminates one term of the recurrence. To use this, first eliminate v_n from the two exact updates:

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \quad (6)$$

Whenever $u_k \equiv 0 \pmod{\ell}$, it follows that

$$-u_{k-1}u_{k+1} \equiv (a_{k-1}u_{k-1})^2 \pmod{\ell}.$$

Thus the negative product of the neighbouring numerators must be a square. Since our numerator is a polynomial, its roots modulo ℓ tell us exactly where to apply this observation.

Let $\ell \nmid 6m$ be prime and let $r \in \mathbb{F}_\ell$ satisfy $f(r) = 0$. The constant $\eta = 6c/m$ is nonzero modulo ℓ , so $r \notin \{0, 1, -1\}$. Choose a sufficiently late k with $k+1 \equiv r \pmod{\ell}$. Then $u_k = \kappa f(k+1) \equiv 0$, whereas

$$f(r-1) = -3r(r-1), \quad f(r+1) = 3r(r+1).$$

Consequently, in $\mathbb{F}_\ell$,

$$(a_{k-1}u_{k-1})^2 = -u_{k-1}u_{k+1} = 9\kappa^2 r^2 (r^2 - 1).$$

Dividing by the nonzero square $9\kappa^2 r^2$ gives

$$\ell \nmid 6m, \quad r \in \mathbb{F}_\ell, \quad f(r) = 0 \quad \implies \quad r^2 - 1 \in (\mathbb{F}_\ell^\times)^2 \quad (\ell \text{ prime}). \quad (7)$$

Every root is covered. A prime at which f has no root imposes no condition.

Let α be a root of f and put $K = \mathbb{Q}(\alpha)$, a cubic field. We claim that $\alpha^2 - 1$ is a square in K . A nonsquare would give a quadratic extension in which an automorphism fixes α but changes the sign of its square root. Chebotarev realises this behaviour modulo a prime: the reduced α lies in the prime field, while the square root does not. That would contradict (7).

Here are the details of this passage. Suppose $\alpha^2 - 1$ is not a square in K , and choose β with $\beta^2 = \alpha^2 - 1$. The nontrivial automorphism of $K(\beta)/K$ fixes α and sends β to $-\beta$. Extend it to an element $\sigma \in \text{Gal}(M/\mathbb{Q})$, where M is a Galois closure of $K(\beta)/\mathbb{Q}$.

Chebotarev's density theorem [3, §3, author-version p. 15] supplies infinitely many unramified rational primes whose Frobenius is conjugate to σ . Choosing a prime of M above each of them makes the Frobenius equal to σ itself. Omit residue characteristic two, primes dividing $6m$, and the finitely many primes where α or β is nonintegral or β reduces to zero. At a remaining prime, Frobenius acts by the ℓ -th power on the residue field, so

$$\bar{\alpha}^\ell = \bar{\alpha}, \quad \bar{\beta}^\ell = -\bar{\beta} \neq \bar{\beta}.$$

The first equality puts $r = \bar{\alpha}$ in $\mathbb{F}_\ell$ and gives $f(r) = 0$. The second keeps both roots $\pm\bar{\beta}$ of $X^2 - (r^2 - 1)$ outside $\mathbb{F}_\ell$. This contradicts (7), which applies to the particular root selected by Frobenius. Removing finitely many primes has left infinitely many choices. Hence there is a $\beta \in K$ with $\beta^2 = \alpha^2 - 1$.

3.4 Traces in the cubic field

To constrain η , we use the square relation to express α through a pair of reciprocal elements in the same cubic field:

$$(\alpha + \beta)(\alpha - \beta) = 1.$$

Put $z = \alpha + \beta$. Then $z^{-1} = \alpha - \beta$ and $2\alpha = z + z^{-1}$. Since $z \in K$ and $\alpha \in \mathbb{Q}(z)$, we have $\mathbb{Q}(z) = K$. Write the minimal polynomial of z as

$$p(T) = T^3 + bT^2 + dT + w, \quad b, d, w \in \mathbb{Q}, \quad w \neq 0.$$

The minimal polynomial of z^{-1} is the monic reciprocal polynomial

$$\frac{T^3 p(T^{-1})}{w} = T^3 + \frac{d}{w}T^2 + \frac{b}{w}T + \frac{1}{w}.$$

These two polynomials give the traces of z and z^{-1} by the same Newton identities. All traces below are from K to $\mathbb{Q}$.

The cubic for α determines its first three traces:

$$\text{Tr}(\alpha) = 0, \quad \text{Tr}(\alpha^2) = 2, \quad \text{Tr}(\alpha^3) = -3\eta.$$

Expressing these same traces through z and its reciprocal gives

$$d = -bw, \quad b^2 + b(w - w^{-1}) = 1, \quad \eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \quad (8)$$

For completeness, the first equality follows from $0 = 2 \operatorname{Tr}(\alpha) = -b - d/w$. After substituting $d = -bw$, Newton's identities give

$$\begin{aligned} \operatorname{Tr}(z^2) + \operatorname{Tr}(z^{-2}) &= 2b^2 + 2b(w - w^{-1}), \\ \operatorname{Tr}(z^3) + \operatorname{Tr}(z^{-3}) &= -3(b^2 + 1)(w + w^{-1}). \end{aligned}$$

The first line is $4 \operatorname{Tr}(\alpha^2) - 6 = 2$. The second is $8 \operatorname{Tr}(\alpha^3) = -24\eta$, because the mixed terms in $(z + z^{-1})^3$ have trace $3 \operatorname{Tr}(z + z^{-1}) = 0$. These give the other two equalities in (8).

Multiplying the middle equation by w and bringing all terms to one side gives $(bw - 1)(b + w) = 0$. Write $w = r/s$ in lowest terms, with $r \neq 0, s > 0$. Substituting $b = -w$ or $b = 1/w$ into the last equation and using $\eta = 6c/m$ gives respectively

$$m = \frac{48c rs^3}{(r^2 + s^2)^2} \quad \text{or} \quad m = \frac{48c r^3 s}{(r^2 + s^2)^2}.$$

Since $\gcd(r^2 + s^2, rs) = 1$ and $m \in \mathbb{Z}$, the square $(r^2 + s^2)^2$ divides 48. Hence $r^2 + s^2$ is 1, 2 or 4. Both r and s are nonzero, which leaves only 2: a sum of two nonzero integer squares cannot equal 1 or 4. Thus $|r| = s = 1$, and positivity of m gives $m = 12$. The only remaining numerators are

$$Q(n) = 2n(n+1)(n+2) + 1 \quad \text{and} \quad Q(n) = 2n(n+1)(n+2) - 1.$$

3.5 The contradiction modulo seven

In the plus case, choose a late block starting at $n \equiv 0 \pmod{7}$; in the minus case, start at $n \equiv 1 \pmod{7}$. The four successive numerators have residues

$$(u_n, u_{n+1}, u_{n+2}, u_{n+3}) \equiv \begin{cases} (1, 6, 0, 2) & (c = 1), \\ (4, 5, 0, 1) & (c = -1). \end{cases}$$

Call the first two denominator residues d_0, d_1 and the displayed numerator residues $c_0, c_1, 0, c_3$. From $0 = a_{n+1}c_1 - d_1$ and $c_3 = -a_{n+1}d_1$ we get $d_1^2 = -c_1c_3 = 2$. Thus $d_1 \in \{3, 4\}$. The first update, however, gives

$$d_1 = a_n d_0 = \frac{d_0(d_0 + c_1)}{c_0}.$$

Completing the square in these two cases gives, in $\mathbb{F}_7$,

$$d_1 + 2 = \begin{cases} (d_0 + 3)^2 & (c = 1), \\ 2(d_0 - 1)^2 & (c = -1). \end{cases}$$

Since $2 = 3^2$ in this field, $d_1 + 2$ must be a square in either case. But $d_1 \in \{3, 4\}$ would give $d_1 + 2 \in \{5, 6\}$, both nonsquares. The middle zero and its two neighbours supplied $d_1^2 =$

2; the preceding update supplied the incompatible second square. All four numerators are used. Both cubics are excluded, proving Theorem 1.2.

Section 2 of the companion proves positive lower density of disagreement with each rational cubic of this form. That lower bound depends on the orbit and the cubic. For Theorem 1.2, exclusion of eventual equality suffices.

4 Scope and verification

The cubic-rate hypothesis permits the extraction of a cubic polynomial from the integer tail numerator. The assumption $a_{n+1}/a_n^2 \rightarrow 1$ in Problem 1.1 alone does not supply that rate, so the general eventual-recurrence question remains open. The [long record's recurrence criteria and further questions](#) retain the other arguments and their precise hypotheses.

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The [verification section of the long record](#) gives the formal correspondence, source revisions and reproduction commands.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 64.
- [2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.
- [3] P. Stevenhagen and H. W. Lenstra, Jr., *Chebotařev and his density theorem*, Math. Intelligencer **18** (1996), no. 2, 26–37, doi:10.1007/BF03027290. Page references are to the linked author version dated 23 March 1995.
- [4] J. Koizumi, *Irrationality of the reciprocal sum of doubly exponential sequences*, Integers **26** (2026), Paper No. A28, 17 pp., doi:10.5281/zenodo.18714404. Locators refer to this published version.
- [5] T. F. Bloom, *Erdős Problem #243*, accessed 28 July 2026.

Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator	Proposition 3.1	Lean · Comparator